chapter 1

INTRODUCTION

Objectives:

After completing this chapter, you can understand the following:

- The objectives of research
- The definition and motivation of research
- Different types of research
- The research approaches and its limitations
- Various steps in the research process
- The criteria of good research
- The ethics and morals in research

From a novice's perspective, research can be defined as the search of knowledge. Oxford dictionary defines research as the systematic investigation and study of materials and sources in order to establish facts and reach new conclusions. Research is pursued within most professions. More than a set of skills, it is a critical way of observing, examining, thinking, questioning and formulating principles that hold true at least for the given space. Almost all professions affirm the need of research either for the advancement of business or for the enlightenment of knowledge. Whatever profession we are in, we ask ourselves a lot of questions for finding new knowledge and ideas. For example, let us consider that you are running a hotel; there are a lot of questions that may help you in increasing your business:

- How many customers do I treat daily?
- Which are the most served dishes?
- Which combo meal is more popular?
- What time does a particular meal hits its maximum order?
- How the customers rate our service?
- What is the average price a customer spends on a dish?

Just by finding answers of these, one can always say that, a very valid investigation has been done for the domain and the results. This will truly help him in making a sure positive progress. This is a very raw example of research that we practice in everyday life. Consider the graph (Fig. 1.1), we have time in *x* axis and knowledge in *y* axis. In our hotel example, there is a single point in *x* attribute, which can trigger the management with a new concept for the advancement of the business. That is, the concept generation point, which is represented as a steep peak.

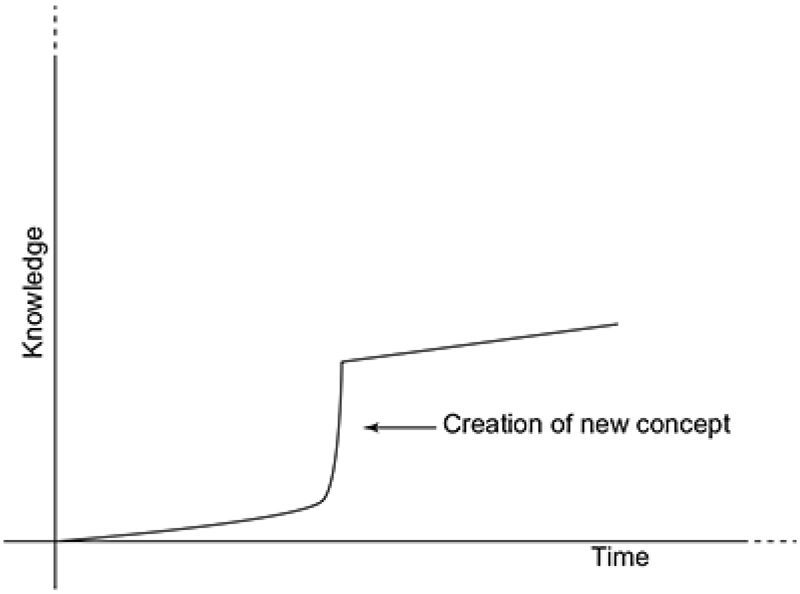

Fig. 1.1 *Concept generation graph*

Redman and Mory defined research as, “a systematized effort to gain new knowledge”. Some people/professionals consider research as a movement, a movement from the known to the unknown. It is actually a voyage of discovery with pleasure and satisfaction. While considering research as an academic activity, it involves a lot of steps such as problem definition, to solve the problem, literature review, data collections, analysis, drawing inference, making hypothesis and arriving at a solution. This book deals each and every activity that a researcher has to perform physically while engaged in research. Research is not just gathering of information from books and other sources. The transportation of knowledge from one form to another will neither constitute a good research. In short and simple, we can define research as, “the systematic process of collecting and analyzing information (data) in order to increase our understanding of the phenomenon about which we are concerned or interested”.

Research is not confined to science, engineering or technology. The research area covers many disciplines such as history, sociology, linguistics and so on. Whatever may be the subject, it discovers, interprets or revises the facts, events, behaviours and theories. The outcome of the research enhances the quality of human life.

Research process is a combination of study, experiment, observation, analysis, design and reasoning. How do we know that cigarette smoking is injurious to health? How do we know that *plasmodium vivax* causes malaria? How do we know that X-Rays can take internal images? All these are outcomes of research. Research provides precise prediction of events, explanation to facts and theories, and various other information to humanity.

Social, scientific and engineering researches focus on mathematical models and theories. The output of an engineering research can be a product, process or even a methodology. Thus, engineering research can be explained in two worlds: explore and develop. Figure 1.2 shows the engineering research hierarchy. Social research is conducted by social scientists through a systematic plan. A quantitative/qualitative dimension is given by these researchers. Social research area includes statistics, political sciences, sociology, media studies and market research. Sampling is one of the key areas in social research process.

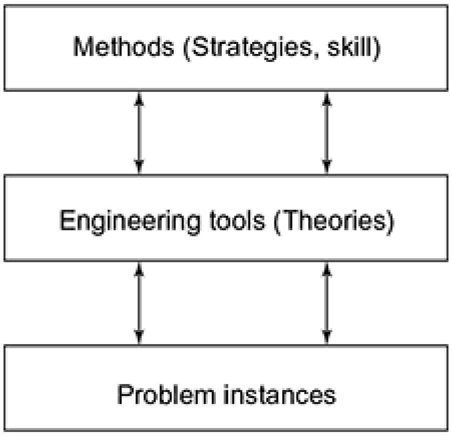

Fig. 1.2 *Research hierarchy*

Scientific research includes those investigations utilized in acquiring new knowledge or correcting and integrating previous knowledge. Basically scientific research forms questions, prepares hypothesis, makes predictions, tests the observations and finally analyses the investigated result. The formal science, physical science, life science, social science, applied science and interdisciplinary sciences are some of the types of scientific research.

1.1 OBJECTIVES OF RESEARCH

It is quite natural to set goals before you start a task. Similarly, setting the goals in your research specifies the objectives related to your work. You can find the answers to all your questions through a series of scientific, recurrent set of queries. Thus, the hidden truth is exposed. So, let us look at how to set the goals of research. Goal setting can be broadly classified into four steps, which is shown in Fig. 1.3.

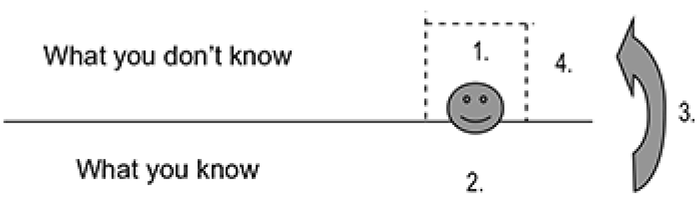

Fig. 1.3 *Setting goals*

- Research objective
- Out of knowledge
- You can do it
- Usefulness

Research objective is the primary aim while carrying out research, where you can have general questions such as what are the aims, and how do they differ from objectives? A statement of what you intend to achieve by undertaking your research is the aim, whereas a statement of what is intended to be achieved at the end of the research is the objective. So, goal setting always starts with the objective, that is, what you intend to find out. Then our focus should be on undertaking some out-of-the-box thinking, which will provide an insight in to how to achieve the set objectives. The usefulness of your research is also a major step while setting the goal of equal or, if possible, more importance should be given to those steps that need be taken to prevent the unhealthy use of your research products. Einstein's famous mass energy equation, $E = MC^2$, was the basis of creating the atom bomb, which is still marked as one of the darkest days in the history of world and is often quoted as an example of this. The impact of such research outcome can destroy the world. Figure 1.4 shows the remains of Hiroshima city, which was destructed completely by the atom bomb. So usefulness and wise use of research outcomes hold a prime role in setting research goal.

Fig. 1.4 *Remains of Hiroshima episode*

Source: old.seattletimes.com

Invention of electricity by Michael Faraday is an example of output of diligent scientific research. But at times unexpected circumstances have led to major discoveries that have changed the course of the world. The discovery of X-rays by William Roentgen is such an example. While conducting a study of electrical rays, he accidentally discovered that a fluorescent covered screen was illuminated by the rays. He also found that the image thus produced could be captured, leading to the world's first X-ray.

Before determining the objective of a research, you should identify the scope of your work. After this, you can determine on what you want to achieve and what decision you want to make. This will help to save time and effort in the later stages of your research. We have to focus on the primary objectives of the research.

- New fact discovery
- Test or verify important facts
- Cause and effect relationship analysis
- New scientific tools, equipment and software, which can solve theories/concepts of scientific and non-scientific problems
- Identification of solutions to scientific and non-scientific problems
- Solve problems in day-to-day life

Even though research objectives are scattered, they can be useful. It helps in identifying and sorting of objectives that help the researcher to specify his/her aim.

Objective of the study should always be realistic, brief but descriptive and be feasible. So, by generalizing these facts, we can summarize the objectives as in the following:

1. To get familiarized with a new theory or an idea by extensive investigation/reading to master each and every concept related to it
2. To generalize the facts that you have studied for a specific class and implement them in a wide population
3. To find the relationships between the various events and factors that can influence the study under focus
4. To draw conclusion by conducting rigorous tests and thereby formulating hypothesis with solid mathematical proof

1.2 DEFINITION AND MOTIVATION

The purpose of the research is to foresee future problems through pursuit of truth as a "global center of excellence for intellectual creativity". The scope of the research is basically defined by the researcher himself.

This helps to prioritize the tasks and even can avoid some issues that are likely to consume much of our time. Try asking yourself the set of following questions before you start up a research.

- What is the purpose of your research?
- What information is being sought?
- How will the information be used?

The answer for these questions is likely to guide you in a right path of research. Research is not just acquiring knowledge; it is the creation of knowledge well around the researcher from which he/she can mine any sort of information. In short, it is not the breadth of knowledge but the depth that is important. The basic purposes for research are to learn something and to gather evidence for any sort of observation or theory. First is to learn something, for your own benefit. Learning is not just restricted to studying theories and formulation of hypothesis, it can be the observation of some techniques that your favourite batsman did while playing a straight drive in cricket. Thus, we may generalize research as an organized learning. Thus, reading an encyclopaedia for the latest innovations and reading a sports' section for last night's game results are both information gathering and, in other words, research.

What you have learned is the source of the background information which you use to communicate with others. In any conversation, you talk about the things you know, and also about the things you have learned. When you write or speak formally, you share what you have learned with others backed up with evidence to show that what you learned is correct. If, however, you have not learned more than your audience already knows, there is nothing for you to share. Thus, with recursive learning, you do your research.

So, it is high time to give a complete definition to the term "research". It refers to the systematic method of solving the problem, formulating hypothesis, collecting the facts, analyzing them and reaching certain conclusions either in the form of solution(s) towards a specific problem or in certain generalizations for some theoretical formulation.

Strive for attaining new knowledge needs self-motivation and creativity. The attitude of the researcher and the commitment that he/she puts into it for the completion of his/her work are really important for having a perfectly designed and well-organized work. There can be a lot of answer for, what makes people to undergo research? The most commonly found answers will be as follows:

- Need a degree with all luxury that it can provide,
- To have that intellectual joy of accomplishing such seemingly exhaustive problems,
- To attain respect in public,
- To serve the public through the new discoveries and improvements of processes,
- To bring about something new in your field that will pave the way for future generations to explore deeper

Not only motivations but there can also be some compelling factors such as some government order, employment conditions and so on. Another important factor that is required for a rigorous research is a guide/mentor who can lead you safe in the sea of research. The guidance and motivation that the mentor provides always lit the fire of motivation in every researcher.

The only way to do great work is to love what you do.

1.2.1 Variables

A variable is any factor that varies in place-time-object context. Variables are inter-connected. A change in a single variable causes simultaneous change in related variables but maintains equilibrium. For example, if we would like to buy a new mobile phone, there are various variables that we look for, such as brand, colour, size, features, operating system and so on. Each attribute is viewed as a variable. Identification of research problem should follow the fixing up of major variable sets as objectives. Figure 1.5 shows the major variables while considering the construction of a new dam. All the variables are separate objectives for the research. The industry, water supply, electricity, agriculture, health issues likewise, all the important factors need to be carefully studied before starting such a project like dam. As mentioned above, all the variables are inter-connected, so the effect of one variable on another and the various factors influencing these variables need to be studied.

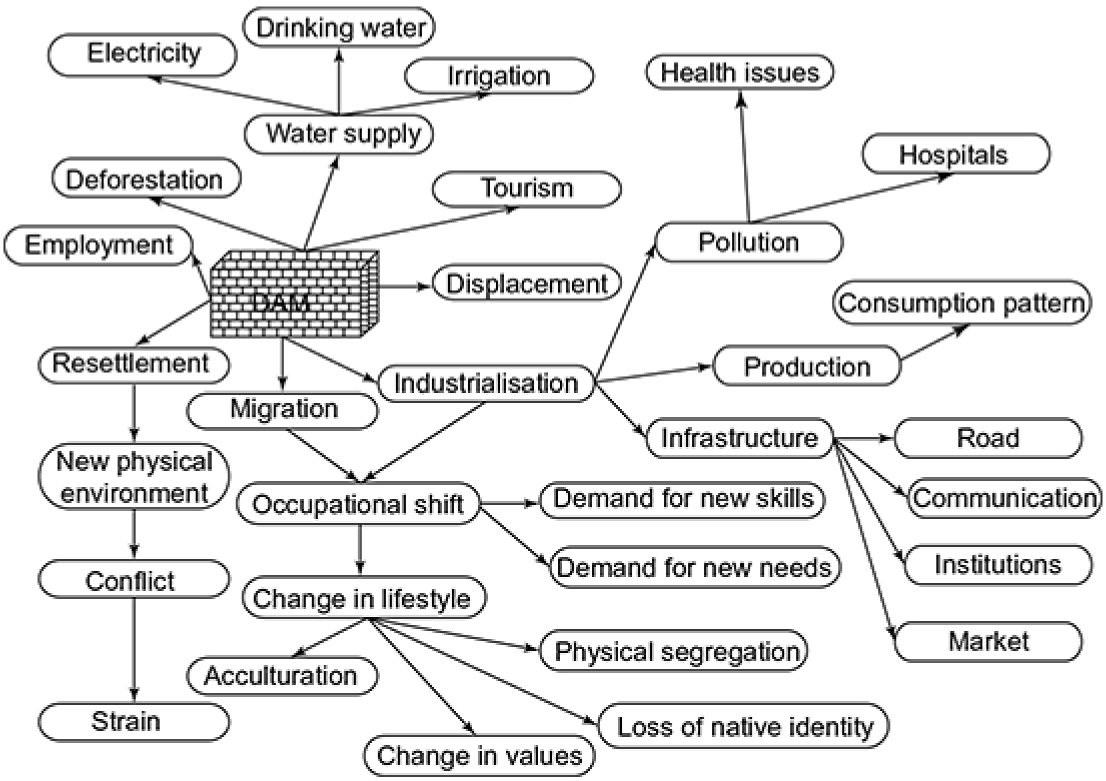

Fig. 1.5 *Variables on a particular research*

There are various types of variables such as independent, dependent, micro, macro, continuous, discrete, qualitative and quantitative. There are certain quantities that are largely dependent on each other. For example, when the salary increases, expense also increases, but the cause of salary increment is an independent variable. Depending on the size, variables are divided in to micro and macro. The annual national income is a macro variable, whereas the annual family income is treated as a micro variable. Depending on the order, variables are divided in to continuous (such as age, income) and discrete (such as religion, occupation). According to measurability, variables are classified as qualitative and quantitative. Attributes such as colour (black, white, etc.) cannot be measured. But variables of unit measures, which can be measured, come under quantitative. Figure 1.6 shows a variety of variables used in research.

Criteria	Types of variable	Example
Effect	Independent	Cause factor
	Dependent	Salary and expense
	Intervening	Social cohesion
Size	Macro	GDP
	Micro	Family income
Order	Continuous	Age, income
	Discrete	Religion (H, C, M)
Measurability	Qualitative	Attributes like black
	Quantitative	Numerical eg. Height

Fig. 1.6 *Types of variables*

1.3 TYPES OF RESEARCH

Research comprises "creative work undertaken on a systematic basis in order to increase the stock of knowledge, including knowledge of man, culture and society, and the use of this stock of knowledge to devise new applications". Research can be classified in many different ways on the basis of the methodology of research. Some of the common research types are as follows:

- Pure research
- Applied research
- Quantitative research
    - Descriptive research
    - Experimental research
- Qualitative research
    - Action research
    - Historic research
- Comparative research
- Exploratory research
- Conceptual and empirical research

Pure research

Pure research is also called as basic or fundamental research. It is an exploratory research done within a pre-described boundary. Mindset and willpower of the researcher is the only way to advance such type of research. Entire work is be driven by the motivation and commitment of the researcher. The work progresses by finding the various variables and their relationship between each other. Pure research always serves as the foundation for applied research. The primary concern of such type of research will be the designing of internal logic and architecture, because of the limited knowledge space. Research concerning some natural phenomenon or relating to pure mathematics or work that tries to generalize some facts are examples of fundamental research.

Applied research

A more glorified form of research is the applied research. Fundamentals that we have come across in pure research are applied to produce some end products. In short, it is an application of theory into practical solutions of problems. Applied researches are basically a group task, trying to create or solve a particular problem. The research becomes truly acceptable when third parties or sponsors use our product with full satisfaction. For example, studying the radio communication channel is a pure research, but creating a product that uses radio communication for any particular task comes under applied research. In the western countries, about 80% of the researches that they carry out are the applied research.

Quantitative research

Quantitative researches are those which can be represented or described according to some numerical system. They are normally associated with large-scale analysis works. These are basically chosen to compare and fine tune the products by considering the amount of output that research process delivers. Statistical tests are being done for measuring the validity and reliability. Experimental and descriptive researches are the major classification of quantitative research.

Descriptive research provides an accurate profile of the group. It describes the process, mechanism or relationship. It gives information and stimulates new explorations. It does not involve any in-depth study, but it reveals just the concepts of theories. For example, while studying the economic activities of a village, one will look for day-to-day activities of the villagers, analysis it and generates a report. There are no deep evaluations, just what he/she perceives reflects in his/her report.

Experimental research measures variations in varied conditions. For example, if we are trying to find the effect of a particular fertilizer, we take a plot and divide it into two halves. In the first half, we keep the fertilizer and the other half will be kept free. During the time of harvest, we can directly see the effect of the fertilizer. Thus, we have a control group and an experimental group. Before and after the true impact study, the research work will be supervised very closely.

Qualitative research

Qualitative research takes an inductive approach. This type of research is found common in social sciences where researchers intend to study social and cultural phenomena. Qualitative research does not have a well-placed definition, but it can be viewed as a work in a “world-view” angle, which has all assumptions but meaning changes from situation to situation. Unlike quantitative no numerical measures are incorporated in qualitative research, instead it uses an in-depth analysis approach by taking case studies or events to study the situations. It is not involved in investigating and developing hypothesis. A common perception of qualitative research is the emphasis on discovery rather than proof. Action research and historic research are the major examples of qualitative research.

Action research makes study while an intervention program is going on. It involves simultaneous intervention and observation or measurement of impact. Historical research explores into historical facts by adopting historical methods. It relies more on the historical documents and other evidences. Studies on the stone encryptions, palm leaf readings, etc., come under historical research. Data collection and data synthesis are the major steps in historic research. The renowned Indian anthropologist R.K. Mukherjee defines qualitative methods as theory generation. Figure 1.7 shows R.K. Mukherjee’s view on research methods.

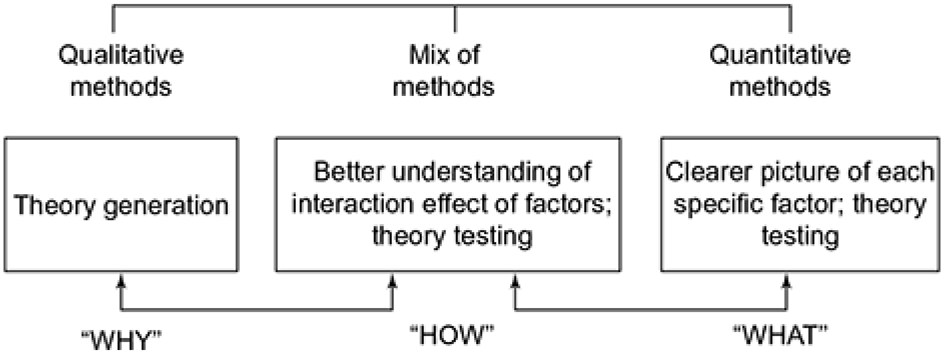

Fig. 1.7 *R.K. Mukherjee's view of research*

Comparative research

It deals with the comparison of similarities or differences under same or varying conditions. For example, study of economic condition of two cities or health condition of people in nearby villages comes under comparative research.

Exploratory research

It is the type of research carried out when the problems are not clearly stated. This research helps in finding the best design and data collection methods. Even though the results are not well defined for making a decision, it can be used for providing insight in to a particular situation. These types of researches cannot be generalized and are not applicable to large populations. Researches on finding the average age of people in a village come under this type of research.

Conceptual and empirical research

Conceptual research is related to some abstract idea(s) or theory. It is generally used by philosophers and thinkers to develop new concepts or to reinterpret existing ones. On the other hand, empirical research relies on experience or observation alone, often without due regard for system and theory. It is data-based research, coming up with conclusions that are capable of being verified by observation or experiment.

1.4 RESEARCH APPROACHES

The various types of research discussed in the previous section enlighten the readers with the two basic research approaches, namely qualitative and quantitative. There can also be more real-time research approaches, but every other approach will be a mixture of either quantitative or qualitative. In a generalized view, we can add logical and participatory approaches along with the aforementioned ones (Fig. 1.8).

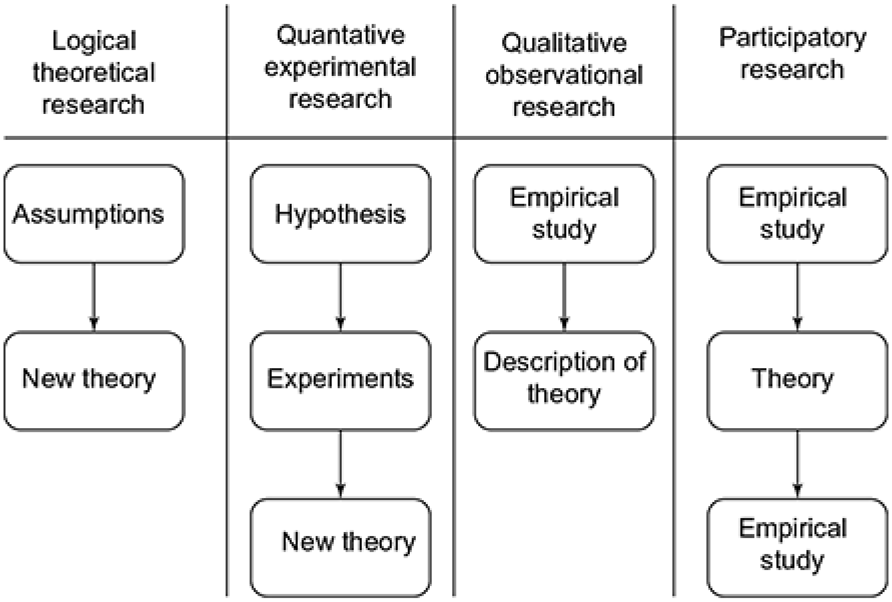

Fig. 1.8 *Research approaches*

There have always been hot conversations and debates on selecting the apt approach for each research problem. But in all researches, a mixed approach can be used to grab the results and to generate the hypothesis. Strauss and Corbin divided the qualitative approaches into three main categories: non-interpretive, interpretive and theory building. Non-interpretive studies are focussed on describing the observations that the researcher has just investigated. Observations need not be analyzed, but can be left for the readers to interpret. Interpretive studies focus on the importance of the analysis to be performed by the researcher. The researchers make out from the empirical observations and create descriptive as well as analytical results. Thus, a "real world" experience is provided to the readers.

1.4.1 Quantitative Research Approach

Quantitative research is generally associated with the collection and conversion of data into numerical form to make statistical calculations from which generalized/specified conclusions are drawn to formulate hypothesis. In a qualitative research, the objective is given prime importance. There may be many hypotheses, but the real question is to select the one that gives the most relevant outcome (result). For this, the researchers make use of the various instruments and materials such as computer, observation check-lists, statistical analysis tools and so on.

There is a reconstruct procedure to accomplish the analysis task; by this, various relationships between variables are found out. Researchers also take good care to control the results and all external factors that can bias the result. This helps to obtain an undisputed final output. The main emphasis of quantitative research is on deductive reasoning that tends to move from the general to the specific and is also known as top-down approach. The validity of the hypothesis is proved by combining one or more valid observations or rules. Like the famous deductive example,

Generalized Statement: All men are mortal.

Example: Socrates is a man.

Specific Conclusion: Socrates is mortal.

Researchers will not have access to the entire population; hence, it is a prime importance to study the sample data (population). Generalizing the results should not be limited to the groups of people but also to the situations. In short, qualitative research approach can be used to develop the understanding required for evaluating if a variable is relevant or not to a given problem situation.

Limitations

- It fails to take account of people's unique ability to interpret their experiences, construct their own meanings and act on these.
- It leads to assumption that facts are true and same for all people any time.
- Quantitative research often produces binary and trivial findings of little consequence due to the restriction on and the controlling of variables.
- It is not totally objective because the researcher is subjectively involved in the very choice of a problem as worthy of investigation and in the interpretation of the results.

1.4.2 Qualitative Research Approach

Quantitative research is about recording, analyzing and understanding the deeper meaning and significance of the revealable variables. The research approach adopted is an inductive method, wherein the researchers develop a theory or look for a pattern on the basis of the data that he/she has collected. It involves a strategic and conventional move from specific to general and is sometimes called the bottom-up approach. There is no pre-determined hypothesis; they will be guided by a set of rules/theories, which provide them the framework to investigate more on the required axis. The data is collected through observations, interviews and through focus groups.

In short, according to the perspective on quantitative research as counting, qualitative research can be seen as proposing which variables to count.

Limitations

- The problem of adequate validity or reliability is a major criticism. Because of the subjective nature of qualitative data and its origin in single contexts, it is difficult to apply conventional standards of reliability and validity.
- Contexts, situations, events, conditions and interactions cannot be replicated to any extent nor any kind of generalizations can be made to a wider context than the one studied with any confidence.
- The time required for data collection, analysis and interpretation is lengthy.
- Researcher's presence has a profound effect on the subjects of study.

1.5 STEPS IN RESEARCH PROCESS

Before explaining the various corners of research, let us have a general idea about how the research process is carried out. There are a lot of feed forwards/back loops within the system in order to attain a complete and fresh research. Figure 1.9 shows the process flow involved in a research process.

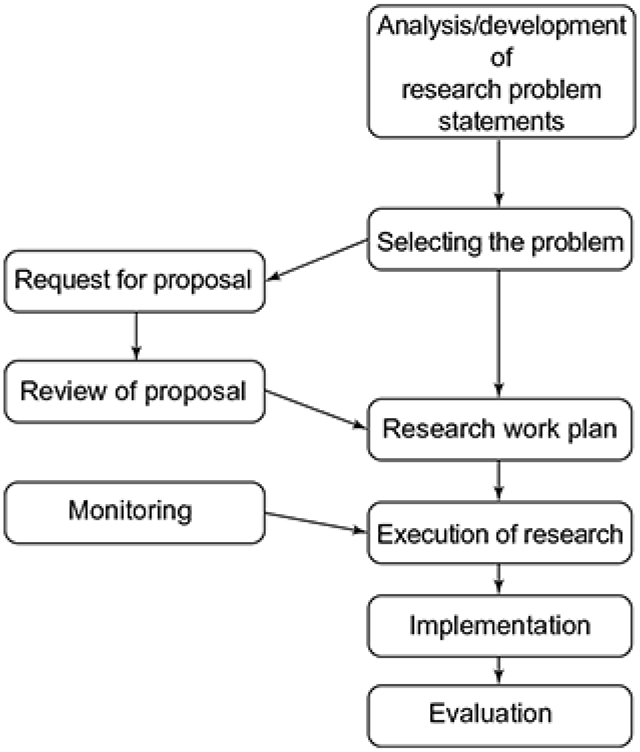

Fig. 1.9 *Research process flow*

We can basically divide research as a six-step process. Figure 1.10 shows the various steps involved in the research process.

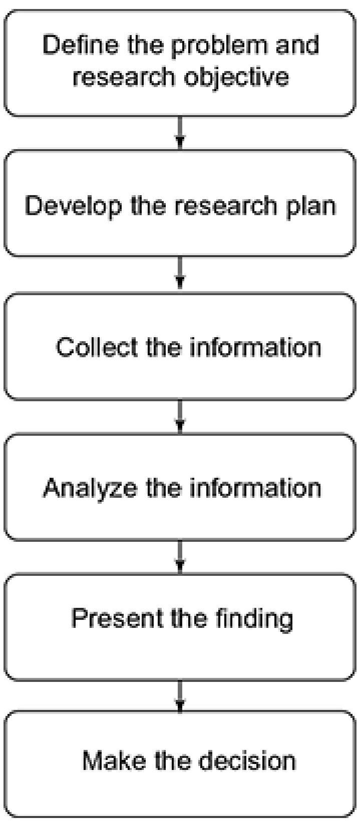

Fig. 1.10 *Steps in research process*

1.5.1 Problem Definition

Defining the problem is the initial step of the research. The researcher must find his/her area of interest and should do a lot of readings in order to find the topic which he/she would like to work with. One major issue during this phase will be the perspective in which the researcher converts the generic area to a specific topic. Researcher needs to be aware of the feasibility of the study and should also look out for the facilities that he/she may get. So when a topic is being focussed, all sort of feasibility study needs to be done. The best way to understand a particular problem is to have discussion with friends or colleagues working in the same area. This will help you to get a wider angle on the area of your interest. We can generally have two types of research: one is association research, where the association of several variables is being studied or those relate to the states of nature. Whichever be the research area, the research should be able to explain the work and area in simple language that shows the ability of researcher to grab deep insight in to the work. Once the researcher feels that he/she is ready with a research area, then he/she should write a sample draft mentioning the aim of his/her work. The synopsis thus written should be reworked and corrected to have a final polished synopsis, which acts as a guide for his/her future work. The synopsis should guide the way through which he/she should work in order to attain the goal. In mean time of research, the goal or objectives can change, so it is always preferred to have a dynamic draft, where in you can make sufficient changes throughout the work. The changes should be made in such a way that the final focus never changes.

1.5.2 Setting Out a Plan

Once the problem is defined, researcher needs to set a plan. A research plan is a thoughtful, compelling and well-written document that outlines your exciting, unique research idea. A typical research plan has four main sections:

- Specific aims
- Significance
- Preliminary studies and progress report
- Research design and methods

The specific aim is a formal statement of the objectives and milestones of a research project. The next section states the research problem including the proposed rationale, current state of knowledge and potential contributions and significance of the research to the field. The preliminary results section describes prior works relevant to the proposed project. They are important to establish the experience and competence of the researcher to pursue the proposed research activity. Purpose of the research design and methodology section is to describe how to carry out the research. This section is critical for demonstrating that the researcher has developed a clear, organized and thoughtful study design.

- It should provide an overview of the proposed design and conceptual framework.
- Study goals should relate to proposed study hypotheses.
- Include details related to specific methodology; explain why the proposed methods are the best to accomplish study goals.
- Describe any novel concepts, approaches, tools or techniques.

1.5.3 Literature Review

Once we have set a plan, it is now time to act according to it. The researcher must learn more about the topic under investigation. To do this, the researcher must review the literature related to the research problem. This step provides foundational knowledge about the problem area. The review of literature also educates the researcher about what studies have been conducted in the past, how these studies were conducted, and the conclusions in the problem area. A good library or resource websites will be a great help to accomplish this task.

1.5.4 Analysis and Hypothesis Formulation

Once the researcher has finished literature survey, he/she should be very clear and should be able to explain the hypothesis behind his/her work. Hypothesis should be very specific and limited to the piece of research in hand because it has to be tested. A simpler mathematical or working model proof should also be able to present to show the consistency of the work. The researcher now analyses the data according to the plan. The data collection methods and related topics of sampling are detailed in Chapter 3. The results of this analysis are then reviewed and summarized in a manner directly related to the research questions.

1.5.5 Presentation and Interpretation

The hypothesis developed and tested now needs to be interpreted to build a theory. The real task of research lies in this area, building a generalized theory from the various facts that has been discovered. This step also deals with, how the proposed system is presented before the audience. It needs to be self-explanatory. The theory and the hypothesis should always provide the audience a feel of freshness with simplicity. The more acceptable work captures more attention. The presentation has an equal importance like interpretation.

1.5.6 Decision Making

Decision making is the final step in research, where the researcher has a very small role. The group to which he/she presents the new theory needs to approve it. The genuine and validity will be questioned by the experts and the researcher need to provide the answers which quench the doubt of experts. A valid and informative piece of research will always be welcomed and finally researcher is rewarded for his/her tedious and hectic work.

1.5.7 Case Study: Sample Research Problem

Now let us look at how we can develop a research problem from an environment around us through the aforementioned steps. Consider a social research on the study of primary child education. In this case, we need to define our problem very clearly. By primary education, are you just looking a population who knows just to read or write, or a population who has completed till a particular class, say standard four, and so on, and also by child which category you are focussing, children below 8, 10 or 14 years. Thus, our problem definition should be apt and clear so that a third person could easily help us or be able to guide us just by hearing our area. After defining the objective, we need to create a work plan, such as from which area are you focussing, what are the sample population, what kind of tools you are using for analysis, how the final result is interpreted and presented, what advantage does this study provides and so on.

The next step is the analysis of the results that are produced from the specified tool. For example, we are using IBM SPSS for the data analysis, we could easily conduct Anova test and get the result. Similarly, if we want to present the data as graph or some charts, the tool has all options. So selecting the tool for analysis also has much important effect. Results after the research should not be just an output that satisfies your work. It should have some impact on the society that helps to create a positive effect on our effort. In this case, after our research analysis and result approval, we should create a huge impact on our result among the society, which will help in uplifting the children, thereby creating more opportunities for their primary education.

1.6 CRITERIA OF GOOD RESEARCH

We have discussed the various types of research, whatever be the type, the research must follow certain guidelines. The purpose of the research should be well defined and the methodologies that we indent to follow should be clearly stated. It helps other researchers who try to repeat the experiments in future for further advancement. Thus, a systematic research, which is well structured and sequentially arranged, will provide more ease to research works.

Another important criterion of good research is the logical flow of ideas. There needs to be a logical reasoning or process of induction/deduction involved while representing the work. The ideas need to be fully supported by well-formed hypothesis or theories. It also gives options for creative thinking for the researcher to arrive at valid conclusions. A good research should always report the type of errors and the flaws in the design. Proper analysis methods and reliability of the results should also be published. The conclusions obtained from the research should be substantiated by valid proofs and the results should be reproducible. Thus, a systematic, logical, replicable research work can be considered as a quality work.

1.6.1 Problems Faced by Researchers

There are various problems faced by researchers who are indulged in dedicated and sincere work. The problems can be either external or internal. Those issues which the researcher have to face from external factors such as co-workers, institute or similar situations can be categorized as external problems. The stress, work pressure and lack of interest due to continuous failure accounts for internal problems. This section details the most common issues found among the researchers.

Lack of support

Researchers may not get proper support from the institution or department where they work. The ambience that the institution offers has a major role in setting the researchers to follow his/her dreams. The support from co-workers, guide and the family is necessary to complete a successful work. The interaction thus becomes a key point.

Lack of funds

Almost all research activities require big financial support. There will always be a funding agency or external agency. It always becomes difficult when the funds get blocked or the agencies withdraw in between. So searching a new sponsor in between is a tedious task. Likewise, inadequate supply of materials that are needed for research can also be a major issue.

Lack of scientific training

The researchers might have no proper knowledge about the research methodologies. A sudden jump into research without proper knowledge may lead to a bad outcome. The ethics, design principles and data collection scheme are well defined in the methodologies, and therefore without a proper knowledge on these may have an ill effect on research.

Lack of study materials

While having a higher level of study, proper books need to be supplied. Libraries may lack sufficient book collections. Also, for online access to top class journals and even the archived copies of works need to be available in the libraries. When there are not adequate materials, literature survey becomes a big question in front of researchers.

Issues in publications

Once we have a valid result or output, we need to publish them in renowned journals. But almost all journals take huge processing time. This can make the researchers restless or even can demotivate them. The review comments from publishers need to be taken by an open heart. It can contain major suggestions, which will be useful to improve your work.

1.7 ETHICS IN RESEARCH

We have read in many books regarding the invention of telescope by Galileo Galilee. But, it is said that a Dutchman, Hans Lippershey, created the first ever telescope, who was denied a patent. Galileo caught wind of this idea and made his own. The controversy regarding this still exists. Here, the idea of telescope was put forward and implemented by Hans, but it was taken from him by Galileo. From a pure theoretical research, it is an ethical issue.

When people hear about ethics, they consider it as a set of rules to differentiate right and wrong. Ethics can also be defined as the norms for conduct that distinguish between acceptable and unacceptable behaviour. Such learning starts at home, school, church or even from the society itself. Consider the society around us; they have legal rules that govern behaviour, but ethical norms take a broader scope and are more informal than laws. Ethical norms are also applicable in research area, to give personal importance to people who conduct scientific or creative activities. Research ethics is the specialized branch, which deals with study of such norms.

Maintaining ethical norms is one of the most important issues to be kept in mind while undertaking research. First and foremost, this helps in defining the principle aims of research. That is, knowledge, truth and avoidance of error bringing down fabrication or falsification of data help in decreasing the possibility of error.

Second, while carrying out research, we need to have the co-operation and co-ordination of several people. They may be from different disciples or even from different institutes. In such situations, we need to have some ethical measures which will help to generate trust, mutual respect and accountability. The various ethical norms such as copyright policies, patents, maintaining confidentiality while reviewing and handling sensitive data are designed to protect the intellectual rights while having collaborative research. Nobody wants to have their idea implemented by others. Thus, ethical norms prevent such acts.

Third, it makes the researcher accountable to the public. For example, while doing research if any misconduct or breaking of some policies or protection is observed, the public can question him/her. There are many laws that prevent unnecessary experiments on animals and birds. As he/she is being funded by the public, he/she is supposed to be answerable to them.

Fourth, ethical norms in research also help to build public support. People are more likely to fund a research project if they can trust the quality and integrity of research. And to gain that trust, ethical norms need to be strictly followed. They also need to promote social values and responsibilities. Research also needs to adhere strictly to the health and safety of the subjects under observation.

For example, a researcher who fabricates data in a clinical trial may harm or even kill patients. Similarly, if a radiologist conducting studies related to UV or IR rays without following the regulations and guidelines may jeopardize his/her health and safety or the health and safety of staff and students.

1.7.1 Morals in Ethics

This section briefs the various characteristics that a good researcher should hold.

Honesty

- Researcher should strive for honesty in all scientific communications.
- Should present only honest report data, results, methods and procedures.
- Fabrication, falsification or misrepresentation of data can lead to many severe issues.
- Should not try to deceive colleagues, funding agencies, public and not even their heart.

Frankness

- Researcher should avoid bias in experimental design, data analysis and data interpretation;
- No personal relationships should come into play while doing peer review, giving grants and even writing testimonies;

- Should make sure that financial or personal interests will never affect research;
- Should have mind to share data, results, ideas, tools and resources;
- Should be open to criticism and new ideas.

Integrity

- Should always keep his/her promises and agreements;
- Should never give false promises and come up with silly excuses when all others are trying hard for his/her success;
- Should be sincere and consistent of thought and action.

Carefulness

- Should always avoid careless errors and negligence;
- Should carefully and critically examine all the work by themselves before giving to peer reviews;
- Should keep a research diary for almost all activities such as data collection, research design and correspondence with guides, friends, agencies or journals.

Respect for intellectual property

- Should honour patents, copyrights and other forms of intellectual property;
- Should not use unpublished data, methods or results without permission;
- Should give credit where it is due;
- Should give proper acknowledgement or credit for all contributors to their research; never plagiarize.

Confidentiality

Sharing data is well acceptable. But when there are situations to handle sensitive data, protect those confidential communications with much care. Papers or grants submitted for publication, personnel records and patient records should not be disclosed.

Responsible

Researchers should always have a commitment to society, should strive to promote social good and prevent or mitigate social harms through research and public education. They should try to improve their self-competence along with the upliftment of the society

Non-discrimination

Discrimination against colleagues and students on the basis of sex, race and ethnicity needs to be avoided. Any factors that are not related to their scientific competence and integrity need not be questioned or challenged.

Subjects protection

When conducting research on human subjects, we need to minimize the harms that may cause. The risk factors need to be well studied and the work needs to be progressed in such a way that it maximizes the benefits. While dealing with vulnerable populations, maximum care needs to be taken such that no actions from their part should make them sensitive. Proper care and respect need to be given to animals while using them in research. They should avoid unnecessary or poorly designed animal experiments

There are various other matters that comes under this category, some of which are mentioned in the following:

- Publishing the same paper in two different journals without informing the editors,
- Not informing a collaborator of their work to file a patent, and thus making sure that he/she is the sole inventor,
- Including a colleague as an author on a paper even though he/she did not make a serious contribution,
- Discussing with their colleague's confidential data from a paper that they are reviewing for a journal,
- Manipulating datasets without providing significant reasons,
- Using an inappropriate statistical technique in order to enhance the significance of research,
- Bypassing the peer review process and announcing the results through a press conference without giving peers adequate information to review the work,
- Conducting a review of the literature that fails to acknowledge the contributions of other people in the field or relevant prior work,
- Stretching the truth on a job application or curriculum vitae,
- Giving the same research project to two graduate students in order to see who can do it faster,
- Overworking, neglecting or exploiting graduate or post-doctoral students,
- Not able to maintain research records periodically,
- Making arrogant comments and personal attacks while reviewing an author's submission,
- Personal grudge while working.

Now we have paved the basic ideas regarding research, the various objectives, motivation, types, approaches, etc. We have also detailed the steps in research process. Now it is high time to have an elaborate study on each of these steps.

EXERCISES

1. Explain the various steps involved in research process.
2. What is scientific research? What is its significance in modern age?
3. Differentiate research methodology from research methods.
4. Explain the various problems that the researchers face.
5. "Research is a dedicated sequential process". Discuss this statement.
6. Differentiate the following:
   - Pure and applied research
   - Conceptual and empirical research
7. Explain the following:
   - Criteria of good research
   - Motivation of good research
8. Motivation and confidence are the keys to success. How can you substantiate this statement?
9. With an example, explain the various steps in research process.
10. Write notes on research ethics.
11. Explain the different criteria in selecting a research objective.
12. Explain the different types of research.
13. What is the relationship of research to science? What do you mean by technology research?
14. What are the different items to be included in research proposal? Explain each in detail.
15. Define and distinguish between theory, law and hypothesis.
16. Explain the ethical considerations related to empirical research.
17. Explain the differences between social, scientific and engineering research.
18. List out the social impacts of research inventions.
19. What are variables? List out the different variables used in a social research.
20. What are qualitative and quantitative methods? How are they useful in research?
21. Explain how scientific collaborations are made while performing research?
22. What are the limitations of quantitative research approach?
23. What is the bottom-up approach in qualitative research?
24. What are the steps involved in the research process for the social/economic surveys?
25. What are the ethical norms in research? Explain the reasons.
26. Explain the ethical principles in research.

27. How animal care is given in research studies? Is it important?
28. How data confidentiality is kept in research process?